

The past, present and future of PCM

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Twenty-five years after its invention, it can be said that pulse code modulation has little past as yet; the real interest is in its future. This future depends a great deal on how well or how badly its main planning problems are tackled during the next decade or so. There is little or no agreed view on the technical and more general points involved in this planning. It is therefore vital that such points should be freely discussed *now*, to try to avoid irreversible wrong decisions.

I am hoping that my own views given in this article will do something to stimulate such discussion — and the fiercer the criticism, the better!

Early progress

Pulse code modulation, or coded step modulation (which I think would have been an apter name), is a good example of an invention that came too early. I conceived the idea in 1937 while working at the Paris laboratories of International Telephone and Telegraph Corporation. When PCM was patented in 1938⁴ and in 1942,⁵ I knew that no tools then existed that could make it economic for general civilian use. Only in the last few years, in this semiconductor age, has its commercial value been felt.

In 1937 I realized, though, that it could be the most powerful tool so far against the effects of interference on speech — especially on long routes with many regenerative repeaters, since these devices could easily be designed and spaced so as to make the noise nearly noncumulative.

The quantizing noise was foreseen, as was the fact that its effects on the listener could not a priori be calculated, since it would be neither constant and additive nor of a nature to produce simple, fixed harmonic content for a given waveform and volume as with simple non-linear distortion. It was clear that nothing but subjective tests could provide the information needed for design; and it is strange that even now some PCM planners and committees seem to place a magical reliance on calculated QN power, before those few subjective tests now in progress have ever been completed, let alone interpreted for the designer. It is strange too that these very necessary, systematic, subjective tests were not started by someone much sooner; it was probably the doubt about the need for PCM itself that delayed them.

My guess of 32 levels for constant speech volume was not far out, although I made a bad, avoidable mistake in not realizing that these levels should be logarithmically companded.

Pulse code modulation was invented mainly for line-of-sight microwave links or link sections, where in 1938 the needed extra bandwidth would have been cheap and easily obtainable, rather than for more limited frequency bands, as in cables, which are now in fact the main fields of application. It is this change of aim for PCM, for quite sound reasons, that has caused most of the technological difficulties so far in its application.

Having had it patented, for understandable reasons I then let the invention slip from my mind until the end of the

war. It was in the United States during World War II that the next step in PCM's progress was made, by Bell Telephone Laboratories. In this important stage, a team under Harold S. Black designed a practical PCM system later produced in quantity for the U. S. Army Signal Corps. Research was also done under Ralph Bown. It is appropriate that this early Bell work should be stressed, for it was the first time that the principles underlying the new system were translated into hardware.

No doubt as the result of their wartime work, Bell Laboratories took the next steps forward. Although suitably cheap, long-life components were still not available, the U.S. Bell System took a long shot, eventually proved correct, in deciding that PCM-steered efforts would be justified at that date as a reasonable bet for future civilian networks.

Apart from patents, the first public disclosures concerning PCM came from Bell Laboratories in 1947.^{6,7} In these articles Goodall and Black described many of the basic principles, and some coder and decoder circuits are given. During the same year, ITT engineers published three papers relating to noise and distortion in pulse count and PCM systems.⁸⁻¹⁰ The next publication¹¹ is a landmark; it is the first account, by Bell Laboratories, of an experimental PCM multichannel link meeting toll quality. The stress was on technical feasibility: in 1948 the economics of the method could be left to the future. In the same issue of the *Bell System Technical Journal* is the first description¹² of the Bell Laboratories' electron-beam coder tube, which in a more sophisticated form is proving useful to this day. In principle it is elegant and simple; but it suffers from the disadvantage of a fairly short life, and needs a feedback loop for sufficiently accurate beam alignment — which problem has been solved, up to a point, quite neatly.

The next article¹³ is also a landmark. In it Shannon, with Oliver and Pierce, explained the philosophy of PCM in terms of previous theoretical work by Nyquist and Hartley but in more usable form, thus making possible a little later his important contributions to information theory, a branch of science vital to telecommunication and computer engineers alike. At the end of that same year, too, Reiling¹⁴ discussed the use of companding in PCM — a point, as I have already said, that might well have been foreseen in 1937.

By December 1948, then, most of the main factors required for efficient long-distance PCM were well realized, together with its advantages and basic theory. The date marks the end of an important phase.

After 1948 there was a growing emphasis on PCM studies in a number of other countries — e.g., Great Britain, France, Japan, and Germany — as well as in the United States. Anything approaching a complete review of it is impossible in this brief article; the examples that follow, chosen partly at random, are intended merely as typical of this world-wide effort.

Though invented by E. M. Deloraine in 1945,¹⁵ a different digital method for speech, now called "delta"

modulation, did not claim serious attention until five or six years later.^{16, 17} As is now well known, the delta method has a basic advantage for the particular nature of speech waveforms in that it has less redundancy than PCM in the effective number of levels per signal-frequency period. On the debit side, however, the coding means is inefficient, because there are only two possible output levels per sample. As will be explained later, there are good grounds for thinking that speech networks of the future may use a combination of PCM and delta. That is why this non-PCM method is included here.

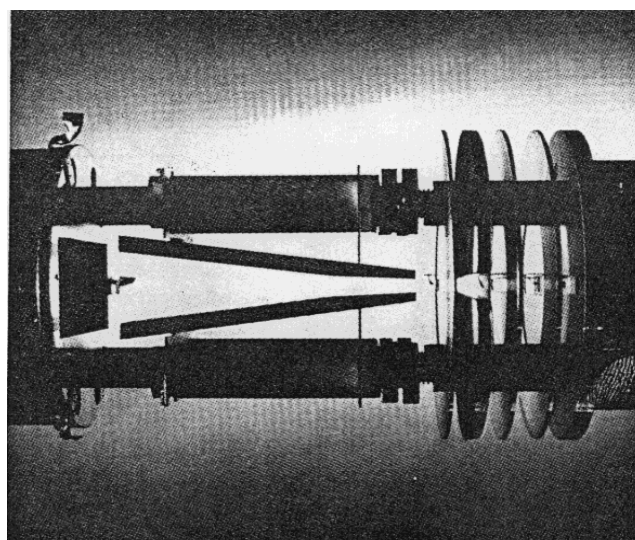
Japan, now one of the most active countries in the PCM field, also started serious studies on the subject during this period, in 1951, at the Electrical Communication Laboratories of the Nippon Telegraph and Telephone Public Corporation. In the first few years the work here and elsewhere in that area included 24-channel systems by coding tubes, together with improved quantized feedback methods to stabilize their beam alignments. The "reflected binary pattern" technique for coding was invented by Dr. Kiyasu during this phase.

After the start of the practical semiconductor era, about 1954, PCM planners and circuit designers began to re-evaluate their projects, for at last suitable tools to justify the relative complexity of the terminal equipment were not only on the horizon but practically within their grasp. Although PCM for civilian uses got off to a good start, progress was nevertheless slow. I think there were two main reasons. First of all, a vast amount of capital was already tied up in the world's existing telephone plant, which at first sight is not readily compatible with PCM. This type of consideration on a number of other occasions has been decisive in delaying improvements as Deloraine pointed out in 1956.¹⁸ The second reason is that at the same time PCM was being developed the older analog methods were themselves being improved. It is always a healthy cold shower, and a challenge, for a pioneer to have to remember that his "wonderful new system" must compete not just with current equipment but with improved versions of the older art at the time his invention is in production — and that if he is too slow he may never catch up at all!

Feedback coding, an older idea, was explored more fully by B. D. Smith in 1953¹⁹; and additionally a digitally companding type was discussed by J. C. H. Davis in 1962.²⁰ The possible advantages, especially in accuracy, of companding digitally within the coder itself rather than externally by a separate device were by then beginning to be realized.

In 1956 an interesting new type of parallel coder, elegant and simple in conception, was invented by A. T. Starr.²¹ It used a square-loop ferrite core, a reliable passive element, as a near approach to a true level-decision device. Experiments showed, however, that because of magnetic breakthrough it would be difficult to get beyond above 32 levels usefully, at present an insufficient number for straight PCM with a practical range of input volume.

Soon after about 1950 it was beginning to be realized that if in electronic exchanges the speech information were in digital form, a number of switching problems could be simplified. A little later it was also realized by PCM



Pictured above is a section of a high-speed PCM coding tube showing the deflection plates (triangular structure) and the electron lens (disks at right). These parts control the focusing of the electron beam. The deflection plates are shaped to provide maximum deflection sensitivity. (Photo courtesy Bell Telephone Laboratories, Inc.)

planners that their new system would initially fit much more easily into the local areas, where the parts had only to be compatible internally, than into the toll routes, where it must interconnect with conventional systems.

Considerations of this kind led Laboratoire Central de Telecommunications, the Paris laboratories of ITT, in 1958 to introduce PCM as a basic feature of their studies into electronic switching methods,²² and led ITT (in their London Standard Telecommunication Laboratories) and the American Telephone and Telegraph Company in their "T1" system, to begin to develop and introduce a self-contained 24-channel method on individual junction-cable pairs into such local areas.^{23, 24} Actual and experimental operational results have been most promising; and, at least for this application, PCM is proving economical already, even without integrated circuitry, which should provide even greater savings.

Since 1961 Japan also has been very active in this PCM short-haul field, mainly on a 24-channel-per-group basis. The rapidity of growth of its industry and the consequent fast increase in telephone demands have made Japan one of the most promising areas for PCM's early application and further development. The first field trials of such a 24-channel system were satisfactorily completed in 1964.

Japan's work during this period also included a method, attributable to Prof. Osatake, for transmitting the code digits on a parallel basis within exchanges, rather than by the serial means more usually employed. It is now being applied to the switching system. Considerable economies are claimed in high-capacity exchanges. In my opinion this version, though specifically foreseen in the first PCM patent,⁴ has been neglected.

In 1963 a new type of digitally companding system was disclosed,²⁵ based on sending the information in two parts, as is done when expressing the characteristic and mantissa

of a logarithm. Also in 1963, what seems to be a new circuit principle was invented, applicable basically to other problems as well as to PCM coding. Called the "equilibrium" method,²⁶ it has potentially for PCM nearly the speed capabilities of a parallel coder and, in addition, it is believed that the circuitry can be still simpler and cheaper than in the serial variety. Improved methods for digital companding by the equilibrium process have also been studied.²⁷

One further coder idea should be mentioned here: it is a many-level time-counting version, believed operable at high bit rates, that does not suffer from the disadvantage of needing a very high speed at the input stage of a binary counter.²⁸ Because time is the coded parameter as in the first coder designed⁴ it is believed that a high degree of linearity can be obtained cheaply in a coder-decoder combination. Such linearity would of course be essential in any PCM equipment that was used for coding an FDM (frequency division multiplex) group or supergroup directly, without prior separation and demodulation.

Much work has also been done in the past 16 years on codes other than simple binary, for two main purposes: (1) to reduce the effects of single-digit errors from any cause and (2) to reduce repeater errors caused by occasional code groups having heavy low-frequency components, in combination with cables or cable pairs in which attenuation falls steeply as the frequency falls. The second of these efforts has led to a number of special codes and coders, examples of which are the alternate mark inversion, ternary type used in the AT&T T1 equipment, and the ITT "low-disparity" code for the same purpose.^{23, 24}

We now come to the progress on a solution which may prove to be a better answer than any other for transmission of speech, whether single-channel or time-shared. This "log differential" method, which is a coded form of delta, was studied first in the United States. Although in principle it is not a new idea, it did not arouse much interest until recently.^{29, 30} Like delta, it can avoid for speech an unnecessary number of transmitted levels; but unlike delta the level number that is still needed — for example, about 32 to 128 — is sent in efficient binary-coded form. Further statistical tests on articulation, naturalness, etc., are essential before we can truly assess the new method, but so far results seem to show that 16 suitably spaced levels on log differential PCM are about equivalent to the direct PCM use of 32 levels at optimum speech volume, and of perhaps a few more where the range of volume is the limiting factor. The resultant saving of at least one transmitted digit is quite important for reducing the transmission bandwidth in most applications. Moreover, the coder is simplified, but not, as first expected, at the expense of the channel equipment.

In theory, the log differential method in its simplest form is equivalent to a network suitably emphasizing the high-frequency speech components, followed by a normal log-law compander and coder. The log differential method has one drawback: it is not suitable for every waveform, only for those of the nature of ordinary speech. If multitone signaling were used on it, for example, we do not yet know whether the resulting inter-modulation terms would be acceptable, although some early tests have shown that they

may be acceptable in normal cases. The digital speech systems of the longer-term future, however, will undoubtedly be efficient in both the signaling and speech paths, and will therefore use digital methods for signaling as well.

Progress also has been made on PCM for television applications. In one case in point³¹ the object was to make television transmission possible on a long-haul waveguide, which almost unavoidably has phase-distortion characteristics, due to many slight discontinuities, that make it unsuitable for nearly every other method. In Japan too a PCM coder for television has been developed. It employs Esaki diodes, with 10-Mc/s sampling and 6-bit coding. In addition, the Bell Laboratories now have a PCM television system in an advanced development state.

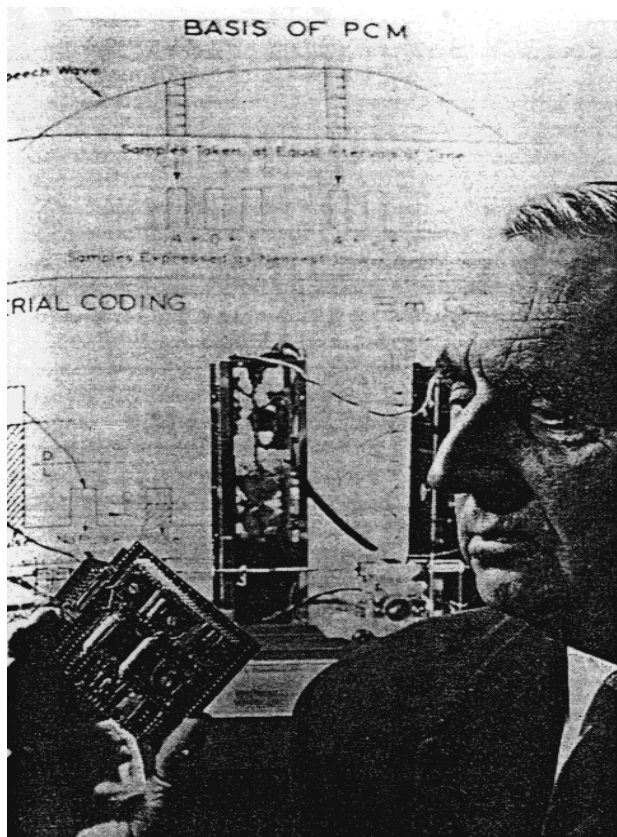
No mention has been made so far of decoders. The reason for their scanty coverage in this article is that they do not have to be decision devices, and therefore their design is basically easy. However, particularly in conjunction with special coding methods — for example, those of digitally companding types — the circuit research worker can still find many interesting problems. To reduce costs, unconventional circuit methods can sometimes be justified. One such unusual design, a revival in new form of an older idea, has been studied by ITT's Standard Telecommunication Laboratories.³²

Present position

Pulse code modulation has been a child with a long infancy; except for certain military uses not described here, in application it is still only in the adolescent stage. As of early 1965 the only PCM systems definitely known to be in regular commercial operation comprise 3000 or so 24-channel groups of the T1 type, all in the United States, and about 12 similar groups designed and used in Italy. Quite a few more T1 trunks, though, are expected to be operating in the United States in the immediate or near future, nearly all of which use pairs in existing cables. Inasmuch as a fair number of such circuits had become full when employed in the usual way and would have been expensive to duplicate, especially in city areas, there was an immediate demand here for a cheaper way of extending the interoffice service. It is most likely that the first sales of commercial-type PCM both in Europe and Japan will also be to meet this kind of demand, probably within the next year or two.

In the immediate future it is probable that the civilian market for PCM equipment will be confined to local-area applications, mainly because the longer routes are in any case becoming cheaper to service by conventional methods, and comprise only 15 per cent of the total global investment in telephone plant. This immediate need for only self-contained types of PCM will perhaps prove beneficial in the long run for the future of the new art, because by these first steps any "teething" troubles can easily be cured without causing widespread inconvenience.

In PCM techniques we have now almost a multitude of coder and decoder ideas to choose from. Much work has been done on repeater design as well, including the problem of timing the regenerative variety. But interterminal



PCM inventor Alec H. Reeves examines a circuit board used on the British Post Office serial interchange telephone system between Guildford and Haslemere, near London, England. The system was designed and built by Standard Telephones and Cables Ltd., British ITT affiliate. Sampling rate is 3000 per second, and the system transmits 23 channels per cable pair.

synchronizing plans, though now being discussed, are by no means completed.

Many national telephone administrations or public utility corporations are showing increasing interest in PCM, and international bodies such as the CCITT are studying it closely; but it will be approximately another five years until the real, relatively wide exploitation of the method will begin.

It is the purpose of the present article to describe the civilian, rather than the military, aspects of PCM. In several countries, however — e.g., the United States — defence departments have studied the subject closely for some years, and have now decided that it will be a major factor in their communication networks.³³

The future

Pulse code modulation is known to work. It is known that its principles are sound. It is known that it has many basic advantages, coupled with some limitations. In the world's networks, will it ever be used on a really large scale? Or will it, except for military and other special applications, remain a mere scientific curiosity? In making our informed guesses on these questions let us start by seeing "the forest" rather than "the trees" — by analyzing the ways, if any, in which PCM principles are likely to be

needed to meet the probable general trends in the world's telecommunication expansion, rather than concentrating on PCM's ability to solve specific, detailed problems. And let us take the unusual course of dealing first with the most distant future date that, except in science fiction, it is sensible to think of, and then the near future, and ending with the middle distance. In that way we can see immediate practical difficulties in better perspective and yet keep our feet sufficiently on the ground. To me the problem is one of boring a tunnel through the next 36-year time span — for I can see the two ends more clearly than I see the middle. In tunneling, it is usual to complete the middle section last.

The long-term view. Let us take the year 2000. Consider first what new factors could then make more efficient methods really necessary in a communications network, not just marginally or even economic to install widely. In a newly developing area, such as many in Africa or Asia, various new systems will have little or no backlog of older methods to link up with or to replace, and for that reason may well come earlier than in the more established areas. Thus, there is a greater freedom of choice in these regions. In already highly industrialized parts of the world, especially where the average man and woman is very telephone minded both for social and business purposes, basically new methods may become almost mandatory if the daily demand for interconnections should exceed a certain figure.

The obvious example to take, then, is the United States. If we extrapolate from the present growth rate and include saturation effects, there should be about 200 million telephone subscribers in the year 2000, compared with the figure of 87.3 million for July 1964.^{34,35} This is not a startling increase. An unforeseen chain reaction between reduced costs and demand could no doubt steepen the curve; so would a still further rise in telephone consciousness than we now expect. But it would be unwise, I think, to assume an extra factor of more than about 50 per cent on these two counts combined. There will no doubt be a great increase in data traffic that could react on the telephone networks; but because some data transmission will use relatively small bandwidths, the total required information capacity will probably not be affected greatly.

By the year 2000 new telephone methods, of which PCM is an example, will by no means be a "must" on a truly wide scale anywhere in the world, although economics may well justify a fairly large mileage by such new and improved means. We shall have to seek another reason, if it exists, for PCM's really large-scale introduction into the world's civilian networks.

In my view, this "other reason" by that date will be the necessity for widespread closed-loop television — a necessity, I repeat, not just the urge for a status symbol that is likely to start this kind of demand in the nearer future. The first need will arise in the field of information retrieval. Consider my own case as an example. It would take me even now about 30 hours in each day of a seven-day week to keep thoroughly up to date in all the scientific and technological subjects that I really need to know in my own sphere of circuitry research alone, if I were to digest and

consider properly all that I read. By A.D. 2000 it would be even more impossible.

Although increased specialization and teamwork will help, there are limits to the usefulness of "knowing more and more about less and less." The only possible answer will be a very greatly streamlined means for getting information, in the form needed and at the exact moments that the needs arise. Every professional man, industrialist, and administrator will require this service.

The only adequate answer will be for a few information-processing centers to be set up in each large industrialized area, staffed by top-grade people, with the information being made available to the public immediately and automatically when a dialed request is made. An ordinary high-speed data link may be adequate for the next 20 years, but by A.D. 2000 the only way to pass the information fast enough to the caller's brain will be to use *moving* pictures. By then a service of this general kind must come, for without it no nation will by modern standards be able even to survive, as the very lifeblood of that survival will be the most efficient use of knowledge.

It is my opinion that a second vital need will arise from the almost impossible transportation problem in the year 2000. Commuters will refuse to accept the delays and inconveniences that even a moderate journey to and from their place of work would entail. Decentralized town planning will alleviate this nuisance, but it is only in light industries that plants and offices can, without undue loss, be sufficiently divided. We shall have to transport the brains, the skills of the staff, not their bodies, to their daily jobs, again involving not merely ordinary data links but a great many private television channels as well. This new service will raise the communication demand in the area, measured in megacycle-miles by several orders of magnitude — which would soon justify a complete modernization of the network to suit it.

But will such a large increase in network bandwidth be technically possible? Yes, but — to make economic sense at all — only by transmitting on optical beams. Some of the needed techniques are not yet with us but, helped by the laser, a fair start has been made,^{36,37} and what we lack now can be available well before the end of the century if the conditions that we must all meet at that date are realized by enough skilled people, and in time. It will happen; only short-term economics make optical methods look relatively unpromising for public use.

What effect will this revolution have on improved digital methods, such as PCM? We have here the reason for including optics in this article, since from basic physics we know that because of the high energy per photon at optical and infrared frequencies the efficiency of optical methods, for a given signal-to-noise ratio, is many times greater when digital rather than analog methods are employed. Pulse code modulation, with between 16 and about 80 levels, would meet many of television's requirements very well; and of course on wavelengths from the visual to the near infrared, the basic bandwidth available is many times the bandwidth that will be required for a long time to come.

In my view, therefore, by the year 2000, pulse code modulation in some form will be the very backbone of the

world's communication systems that are internal to national or still larger units. But except for satellite routes, the widespread transoceanic use of PCM may have to wait until later, because the severe technological problems inherent in an optical submarine cable may not be solved until some years further ahead.

It is premature to make any predictions as to the PCM equipment that will probably be used at that time; even now there are many designs, and principles, to choose from. A vital point, though, is the means used to line up the gate-opening timings at the many regenerative repeaters, terminals, and drop-off equipments, a matter that is presently being debated. One school of thought favors a "start-stop" method at the distribution points, with storage and retiming where necessary; another sees this answer as one that could mortgage future good planning for the convenience of only short-term goals. Although the arguments are fairly well balanced, I share the second view. I believe that with the right principles and devices a network truly synchronous in average frequency, that can also operate as independent local units in emergencies, is not only feasible technically but economically justified at a fairly early stage.³⁸

The next 12 years. In the period immediately ahead PCM is likely to continue at first from a relatively small number of nucleation centers, and then to show accelerating growth. In the United States the Bell T1 system and its successors have a promising early future, especially for the "extended area" type of application with two separate charging rates that has been so far one of the main factors in the T1's early success. It can be expected that similar systems, for example the ITT version,²³ will be installed within the next two years in Europe; and in Japan there will be a short-haul system operating by the end of 1965. In France we can foresee in 1966 a trial switching system for local areas, probably decentralized to concentrators and using PCM throughout. If these systems are successful, other countries will no doubt follow suit.

It is likely that by 1968 both Europe and the United States will start to use PCM for tandem-exchange working, with integrated switching and transmission by PCM being fully operational on an experimental basis in some urban and suburban areas by 1969.

As to long-haul PCM systems, true operational trials will probably be delayed until a little after this 12-year period, the work in this time span being confined to extended planning and apparatus development. On the longer routes the relative economics for the next 12 years or so of FDM versus digital schemes such as PCM is at present a very controversial issue. One authoritative view from Bell Laboratories is that digital methods will be cheaper if a facility is provided for carrying mixed traffic. A 2000-channel digital pipe, using from 200 to 300 megabits per second, is foreseen in these quarters.

For the very heavy real-time traffic in terms of bits per second per mile that I have predicted for the year 2000 there is no doubt that optical PCM methods will have to be used; but for shorter-term comparisons many opposing factors must be considered.

The need to provide for mixed traffic on some toll routes for perhaps a fairly long time raises a rather severe technical problem that is both interesting and challenging. For efficiency and economy, demodulation into separate channels followed by time-shared PCM must be avoided; but to code the whole FDM group into PCM directly requires a degree of linearity in the coder of a much higher order than in anything so far needed in practice. I think it unlikely that the "tour de force" method, by extending present coder designs to their very limits by good engineering, can give the cheapest long-term answer; past history is against this kind of approach. The Karbowski-Craven variety of time-counting coder²⁸ is probably sounder. We can keep in reserve an additional principle, "stored negative feedback,"³⁹ for the most difficult cases.

In at least the next 12-year period priority needs for widespread, efficient, secure defense networks will give a large extra stimulus to digital systems, which will undoubtedly help to speed up the more general applications of PCM. In spite of some obvious arguments to the contrary, satellite planners too are beginning to press for digital speech, since such methods may prove more suitable than analog for satellite sharing by more than one pair of countries.

In Japan it is planned to connect the industrial centers by toll paths using PCM on free-space microwave links, each equipped with 120 channels. The target date is well before 1976.

The middle period (1976 to 1988). By 1976 or so, many PCM-type interexchange trunks will be operating, both in the United States and elsewhere. In all major countries we can expect to see in operation at least a few complete PCM local-area networks, as well as a few toll routes by new, digital cables. A civilian PCM international network, however, is unlikely until the end of this period, largely because of delays in reaching the necessary political and technical agreements.

Toward the end of the phase the United States will probably have installed a fairly long experimental section of optical pipe. There may be some rather short free-space laser-beam trunks operating between the tops of city buildings, with or without added towers. The economics of PCM for combined transmission and switching will have been proved.

Telephone signaling, now still lagging behind current progress on the speech paths, will probably have nearly made up the lost ground by the end of this period. The technical advantage of digital signaling will be clear to all long before 1988; it is the present investment in the older plant and ideas that will set the changeover dates.

Conclusion

We have seen that PCM has had a slow growth. We have examined some of the salient points in its early stages and in present-day PCM thinking and trends.

Crystal balls do not usefully mix, so I have looked through just one of them, my own, to describe what I see of the general communication landscape in the year 2000 and, in particular, of PCM's place in it — to my eye a major place in a wider digital background. The scenery as a whole

can be ignored only at the peril of any nation which at that date wishes to survive even by today's standards. If this picture is deemed wrong, so be it; but if it is thought right in its essentials, let us do our utmost to avoid at least the major kind of mistake that has so often been made in the past, the lack of adequate foresight in planning, national and international, that can hang a millstone around our necks that may take decades to remove.

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